

Chapter 35 — Urologic Evaluation of the Child — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Suspected cord torsion** → no imaging → immediate exploration + contralateral orchiopexy.
- **TWIST score**: 0 no ultrasound (US); 1–5 US; ≥6 straight to theatre.
- **Salt-wasting CAH** (congenital adrenal hyperplasia) crisis day 4–15 → don't discharge early.
- **Infected obstructed kidney** → prompt stent or nephrostomy.
- **Children hide shock**: hypotension = late, decompensated.
- **Bladder capacity (mL)** = (age + 2) × 30.
- **Suspected urethral injury** → no catheter → RUG (retrograde urethrogram); fails → suprapubic tube.
- **Postnatal US after 48 h** — immediately if bladder outlet obstruction suspected.
- **Febrile infant <24 mo**: catheter/suprapubic aspirate (SPA) urine; bag → false positives.
- **Undescended testis**: ultrasound not recommended.

TRIAGE — EMERGENT REFERRALS

- **Acuity tiers**: emergent; urgent <24 h; semiurgent 1–3 wk; routine.
- **Commonest acute scrotum**: appendage torsion — prepubertal, supportive care.
- **Appendage torsion on US**: epididymal hyperaemia, appendage rarely seen.
- **Cord torsion**: ≈1 in 4000 males; most >10 years.
- **TWIST (max 7)**: swelling 2, hard testis 2; absent cremasteric, nausea/vomiting, high-riding 1 each.
- **Viability** inversely related to torsion duration.
- **Manual detorsion** never replaces exploration.
- **Neonatal (extravaginal) torsion**: firm painless testis; nubbin; urgent exploration preferred.
- **Commonest cause ambiguous genitalia**: CAH; 21-hydroxylase ≈95%.
- **CAH tests**: 17-hydroxyprogesterone + electrolytes; screen may return post-discharge.
- **Salt-wasting death**: hyperkalaemia, dehydration, shock.
- **CAH treatment**: lifelong gluco- + mineralocorticoid.
- **Exstrophy**: ≈2.15/100,000; cord tied 2-0 silk (no clamp); cling film.
- **Exstrophy** → immediate transfer to paediatric centre.
- **Commonest infected obstruction**: obstructing stone.
- **PUV (posterior urethral valves) outlet obstruction** → vesicostomy if needed.
- **Renal vein thrombosis (RVT) triad**: haematuria + flank mass + thrombocytopenia.
- **RVT**: male, left, two-thirds in first 3 days; Doppler US diagnosis.
- **Paraphimosis**: ask circumcision status; exclude hair tourniquet.
- **Paraphimosis adjuncts**: ice 5 min; Coban 15–30 min; sugar 1–2 h.
- **Failed manual reduction** → dorsal slit/circumcision under anaesthesia.
- **Priapism**: ≥4 h, unrelated to stimulation.
- **Commonest paediatric priapism**: ischaemic; commonest cause sickle cell disease.
- **Ischaemic gas**: pO₂ <40 mm Hg, hypercapnia, acidosis, glucopenia.
- **Sickle cell priapism**: oxygen + intravenous fluids; no cold packs.
- **Nonischaemic**: trauma, painless → initially expectant.
- **Neonatal priapism**: benign, painless; monitor up to 5 days.
- **Commonest injured genitourinary organ**: kidney; blunt mechanism.
- **AUA (American Urological Association) 2020 imaging**: haematuria + systolic BP (SBP) <90, or worrying mechanism/signs.

- **Renal imaging**: CT (computed tomography) preferred; US acceptable; AAST OIS (American Association for the Surgery of Trauma Organ Injury Scale) as adults.
- **Renal trauma default**: nonoperative; else stent/drain/angioembolisation.
- **Abuse work-up in straddle injury**: <9 months, hymenal/vaginal/perianal, severe, history mismatch.
- **Female genital trauma**: ≈80% straddle; urethral injury ≈5% of pelvic fractures.
- **Female urethral injury**: examination under anaesthesia + cystoscopy; suprapubic tube, delayed repair.
- **Male scrotal haematoma** → US to exclude rupture.
- **Toilet-seat crush**: mean ≈3.9 yr; usually expectant.
- **Retention causes**: drugs, UTI/balanoposthitis, constipation, neurogenic, mass, trauma.
- **Retention without catheter**: analgesia, sitz bath, disimpaction.

TRIAGE — URGENT, SEMIURGENT, ROUTINE

- **Prenatal hydronephrosis**: APD (anteroposterior diameter) >7 mm third trimester or UTD (urinary tract dilation) A2–3.
- **Pre-discharge US if**: solitary kidney, bilateral, or outlet obstruction.
- **Neonatal creatinine** = maternal first 48 h; follow downward trend.
- **Male + bilateral hydronephrosis/thick bladder** → VCUG (voiding cystourethrogram) before discharge.
- **Most prenatal hydronephrosis** needs no neonatal intervention; close follow-up.
- **Suspected outlet obstruction** → urethral feeding tube/catheter.
- **Tea-coloured + casts + protein** → glomerular: C3, C4, antinuclear antibody, antistreptolysin O.
- **Isolated terminal haematuria** → urethrorrhagia/stricture.
- **Post-bone marrow transplant clots**: BK virus, cytomegalovirus, adenovirus cystitis.
- **Persistent gross haematuria, normal imaging** → cystoscopy + bilateral retrograde pyelograms.
- **Wilms**: commonest renal cancer <15 yr; peak 1–5; ≈10% syndromic.
- **Renal tumour**: firm non-tender mass, rarely crosses midline.
- **Neonatal renal mass**: congenital mesoblastic nephroma.
- **Sickle cell trait + renal mass**: renal medullary carcinoma.
- **Testis tumour**: ≈1% solid tumours; peaks <3 yr, postpubertal.
- **Prepubertal testis mass** usually benign; commonest malignant yolk sac.
- **Testis markers**: α-fetoprotein, β-hCG (human chorionic gonadotrophin), lactate dehydrogenase; precocity → Leydig cell.
- **Leydig suspicion**: add testosterone, follicle-stimulating and luteinising hormones, prolactin.

- **Primary amenorrhoea:** no menses by 16 (normal development) or 14 (none).
- **Cyclic pain + amenorrhoea** → outflow obstruction (septum, imperforate hymen).
- **Interlabial mass first step:** exclude upstream obstruction by ultrasound.
- **Older-girl interlabial mass:** add urethral prolapse.
- **Hernia/communicating hydrocele:** operate on examination or compelling history.
- **Communicating hydrocele:** fluctuates; smaller in morning.
- **Hernia precautions:** pain, firm non-reducible bulge, erythema, inconsolable infant → emergent.
- **Operation:** high ligation of processus at internal ring.
- **Empty hemiscrotum at birth:** undescended vs in utero torsion.
- **Descended at birth, empty later:** retractile vs ascending testis.
- **Varicocele:** ≈15% adolescents; Lambert volume = $L \times W \times H \times 0.71$.
- **Non-decompressing supine/right varicocele** → retroperitoneal US.
- **Physiological phimosis:** ≈1% at 17 yr; never forcibly retract.
- **Treat asymptomatic phimosis** if vesicoureteral reflux (VUR)/urinary tract infection (UTI); steroids if circumcision declined.
- **BXO** (balanitis xerotica obliterans) with phimosis → treat.
- **Meatal stenosis:** circumcised; meatus <5 Fr; dorsal deflection → meatotomy.
- **Labial adhesions:** peak 13–23 mo; asymptomatic → observe; recurrence 7–55%.
- **Labial examination:** frog-leg; pull labia straight out, not laterally.
- **Microhaematuria:** >5 red cells/high-power field (RBC/hpf), confirm microscopy, exclude UTI.

EVALUATION OF THE PAEDIATRIC PATIENT

- **Consent** from guardian; **assent** from child.
- **Respiratory rate** shouldn't stay >60; heart rate ≈ 2–3 × respiratory rate.
- **Lower-limit SBP:** 60 neonate; 70 to 1 yr; $70 + 2 \times \text{age}$ to 10; 90 after.
- **BP cuff ≈ 2/3 arm:** small → falsely high; large → falsely low.
- **Shock first sign:** tachypnoea, tachycardia.
- **Body mass index (BMI) percentiles ≥2 yr:** underweight <5th; overweight 85–95th; obese ≥95th.
- **Tanner II (boys):** testicular/scrotal enlargement first; V = hair to thighs.
- **Micropenis:** stretched length <mean –2.5 SD (term ≈1.9; adult 9.3 cm).
- **Stretched length:** pubic symphysis → glans tip.
- **Hard-to-feel testis:** frog-leg; cross-legged or catcher's squat.
- **Simple dimple:** in cleft, <5 mm, base visible → no imaging.
- **Image:** palpable lipoma, tag >5 mm, deep/atypical dimple, deviated crease.
- **AAP (American Academy of Pediatrics) UTI:** pyuria/bacteriuria + ≥50,000 colony-forming units/mL, catheter/SPA.
- **Microhaematuria:** 0.5–3%; commonest thin basement membrane, IgA nephropathy, hypercalciuria.

- **Dipstick blood false positive:** haemoglobinuria, myoglobinuria → confirm microscopy.
- **Microhaematuria threshold:** child >5 vs adult ≥3 RBC/hpf.
- **Persistent microhaematuria** → repeat 2–3 wk → imaging + calcium/creatinine (Ca/Cr).
- **Ca/Cr abnormal:** >0.4 age 2–5; >0.2 age >5.
- **Protein/Cr normal:** <0.7 yr 1; <0.4 to 4 yr; <0.2 >5 yr.
- **Hypercalciuria:** fluids + modest sodium restriction; renal US.
- **Proteinuria:** first-morning sample (orthostatic common); AAP: don't screen.
- **Nephrotic range:** protein/Cr >3.0.
- **Persistent non-orthostatic proteinuria** (2 morning samples) → nephrology.
- **Specific gravity:** range 1.002–1.040; <1.010 dilute/diabetes insipidus; >1.020 dehydration.
- **Chronic kidney disease anaemia** once glomerular filtration rate (GFR) <43 mL/min/1.73 m².
- **Ideal GFR marker:** filtered, not secreted or reabsorbed.
- **Imaging principle:** ALARA (as low as reasonably achievable).
- **MAG-3** (mercaptoacetyltriglycine) = obstruction + split function; **DMSA** (dimercaptosuccinic acid) = scarring + split function.
- **Radiation-free:** ultrasound, magnetic resonance imaging (needs sedation).

CHILD COMFORT

- **Tablet distraction ≈ midazolam** for preoperative anxiety.
- **Midazolam before VCUG:** less recall, results unaffected.
- **EMLA 30–45 min** before office procedures → avoids general anaesthesia.
- **LMX** = lidocaine 4%; **EMLA** = lidocaine 2.5% + prilocaine 2.5%.
- **Lidocaine gel for catheterisation:** mixed results.

CLASSIC TRAPS

- Trap: cremasteric reflex present → not torsion — Actually: presence does not exclude torsion.
- Trap: scan before exploring suspected torsion — Actually: clinical diagnosis; imaging delays.
- Trap: normal newborn screen excludes CAH — Actually: result may be late; only 21-hydroxylase screened.
- Trap: normal BP → no shock — Actually: hypotension signals decompensated shock.
- Trap: cold packs for all priapism — Actually: can induce priapism in sickle cell.
- Trap: haematuria/tachycardia reliably flag paediatric renal injury — Actually: mechanism predicts better.
- Trap: day-1 ultrasound for hydronephrosis — Actually: after 48 h (underestimates earlier).
- Trap: day-1 creatinine reflects neonatal function — Actually: maternal for 48 h.
- Trap: bag urine for febrile infant UTI — Actually: catheter/SPA; bag false positives.
- Trap: ultrasound locates undescended testis — Actually: poor sensitivity/specificity; not recommended.